SANTHOSH JARUPLA

Senior Software Engineer | 5G NR gNB Stack | L2 | MAC | RLC | PDCP
santhoshjarupla12@gmail.com | +91 7661012187 | linkedin.com/in/santhosh-jarupla-909b74a4 | Hyderabad, India

PROFESSIONAL SUMMARY

Wireless Software Engineer with M.Tech in Communication Systems (NIT Trichy) and 6+ years of industry experience in 5G NR gNB protocol stack development. Core expertise spans L2 stack feature development (MAC, RLC), MIMO, Carrier Aggregation, and 3GPP NR specification analysis — all within commercial gNB systems for Ericsson. Currently working at Saab India Pvt. Ltd. (for Ericsson). Seeking to leverage deep gNB stack experience and wireless fundamentals.

KEY SKILLS & TECHNICAL COMPETENCIES

Wireless Technologies: 5G NR (Sub-6 GHz, SA/NSA), LTE, LTE-Advanced
gNB Stack (Hands-on): MAC, RLC, L2 Protocol Stack, Carrier Aggregation (6CC), MIMO (2x2, 4x4)
Programming: C/C++ (proficient), Python (scripting)
Tools & Debugging: GDB, Wireshark, QXDM, EMCA, TM500 Simulator
Standards & Methods: 3GPP 5G NR Specifications (TS 38.xxx), Agile, Git, ClearCase

PROFESSIONAL EXPERIENCE

Senior Software Engineer – gNB NR | Saab India Pvt. Ltd. | Client: Ericsson *Dec 2025 – Present*

- **Developed the EOS (Energy Optimised Scheduling) feature** for PDSCH allocation in 5G NR SA, optimising resource allocation to reduce gNB energy consumption while maintaining scheduling performance.
- Implemented **PDCCH power boost for MSG2**, enhancing Random Access response reliability by improving PDCCH link performance during the RA procedure.
- Contributing to **modem system design documentation** and internal architecture studies, supporting L2 stack design and behaviour alignment.

Senior Software Engineer – L2 (gNB) | GlobalLogic | Client: Ericsson *Sep 2024 – Dec 2025*

- Spearheaded **5G NR gNB feature development** by implementing an Audit mechanism in **C/C++** to clean up dangling UE sessions from external SCells, enhancing **L2 protocol stack** stability and resource management.
- Contributed to **5G NR IODT** (Inter-Operability and Design Testing) by analysing **3GPP NR specifications**, identifying a critical L2 bit capacity constraint for **6CC Carrier Aggregation** that impacted gNB system design.
- Owned design, development, and execution of **unit tests** (UE-CE, DDT, ECE) for gNB L2 components; automated ECE test case generation to accelerate feature validation cycles.
- Worked closely with Ericsson system architects and test teams to **debug gNB stack issues**, correlating L2 logs for root-cause analysis.

Engineer II, R&D – L2 (gNB) | CommScope *Oct 2022 – Dec 2023*

- Executed **system-level feature coding** in **C/C++** for **5G NR gNB protocol stack**, implementing NSA-TDD 40 MHz call configurations across MAC and RLC layers.
- Designed and developed mandatory cell setup parameters for NSA-TDD 40 MHz, ensuring full compliance with **3GPP 5G NR specifications** and gNB interoperability requirements.
- Directed OTA regression testing (NSA-TDD) and unit tests for **4x4/2x2 MIMO** configurations in SA/NSA modes, validating gNB release stability.
- Conducted service interruption testing (ARFCN, PCI, BW changes) to identify and resolve potential gNB stack issues, ensuring robust **wireless communication** performance.
- Developed **L2 CLIs** for test instrumentation, enabling efficient debugging and validation of gNB protocol stack test harnesses.

Engineer I, R&D – L2 (gNB) | CommScope *Aug 2020 – Oct 2022*

- Implemented 500+ **Operation & Maintenance (OM) parameters** for the CATMI feature across the **5G NR gNB L2 stack** (MAC, RLC), directly supporting **4x4 MIMO** feature delivery.
- Conducted regression and **unit testing** with the TM500 simulator, validating gNB protocol stack performance for 5G NR and LTE configurations.
- Enhanced **C/C++ code robustness** by resolving Coverity defects including memory leaks, buffer overflows, and null pointer dereferences, improving gNB modem software reliability.
- Debugged complex integration and SVT defects using **GDB, Wireshark, and QXDM**, resolving complex L2 stack issues within live gNB environments.
- Established LTE OTA setups and developed unit tests for UE attach and data throughput, confirming end-to-end gNB system sanity.

Trainee Engineer – L2 | CommScope *Aug 2019 – Jul 2020*

- Developed **Wireshark dissectors** in **C/C++** to decode MAC interface messages in the gNB stack, accelerating L1/L2 debugging workflows.
- Performed sanity and unit testing for CatM1 (IoT) feature development, validating UE attach and data throughput on Phybypass setups.
- Created CLIs to display OM values and cell configuration parameters, supporting **gNB L2 debugging** and system analysis.

EDUCATION

M.Tech – Communication Systems | National Institute of Technology (NIT), Trichy *CGPA: 7.71*

Relevant Coursework: Wireless Communications, Communication Theory, Digital Communications, Data Networks